

Tutorial: Probability and Statistics (MCC505)
Monsoon Semester: 22-23

1. Suppose the average grade in a quiz is 70%. Give an upper bound of the proportion of students who scored atleast *90%*.
2. Suppose a coin is flipped 20 times. Find a bound for the probability that head will come atleast 16 times. *Take $P(H) = 0.2$*
3. Suppose a fair coin is flipped 100 times. Find a bound of the probability that number of head will be atleast 60 or atmost 40.
4. Suppose someone gives you a coin and claims that this coin is biased; that it lands on heads only 48% of the time. You decide to test the coin for yourself. If you want to be 95% confident that this coin is indeed biased, how many times must you flip the coin? You can allow a precision error of 1%.
5. Let $X_1, X_2, \dots, X_n$ be i.i.d. $U(0, 1)$. Show that $Z_n \xrightarrow{P} c$ where c is a constant. Find c . *where $Z_n = (X_1^2 + X_2^2 + \dots + X_n^2)/n$*
6. A computer has a random number generator that generates number between 0 and 1. Suppose we run it 100 times. Find $P(S_{100} \geq 52)$ applying CLT. Here S_{100} denotes the sum of first 100 generated numbers.
7. By clinical study, it was found that people over age 50 had a mean cholesterol 54 and sd 17. Suppose a physician has 40 patients over age 50 and wants to determine the probability that the mean cholesterol of 40 patients are 60 or more. (Apply CLT).
8. Let $\{X_n\}$ be an i.i.d. sequence of standard normals. Let $Y_n = \frac{1}{n} \sum_{j=1}^n X_j^2$. Use the CLT to find n so that $P(Y_n \geq 1.01) = 5\%$.
9. (a). Let probability density of a random variable X be $f(x) = \begin{cases} \frac{5}{x^6} & \text{for } x \geq 1, \\ 0 & \text{otherwise.} \end{cases}$
What bound does Chebyshev's inequality give for the probability $P(X \geq 2.5)$? For what value of α can we say $P(X \geq \alpha) \leq 15\%$?

(b). Let $\{X_n\}$ be a sequence of random variables having common mean 5 and common variance 2. If the covariance between any two random variables in this sequence is negative, prove that Weak Law of Large Number holds for $\{X_n\}$.

Tutorial Solution

Chebychev's inequality

1. Markov Inequality

$$P(X \geq a) \leq \frac{E(X)}{a}, \quad \text{Given } E(X) = 70\%$$

$$P(X \geq 90) \leq \frac{70}{90} = \frac{7}{9} = 77.8\% \text{ Ans.}$$

2.

$$P(H) = 0.2 \quad n = 20, \quad E(X) = np = 20 \times 0.2 = 4$$

$$P(X \geq 16) = \frac{E(X)}{16} = \frac{4}{16} = \frac{1}{4} = 0.25$$

$$P(X \geq 16) = 0.25 \quad (\text{using Markov Inequality})$$

using the binomial distⁿ, we get.

$$P(X \geq 16) = \sum_{x=16}^{20} \binom{20}{x} (0.2)^x (0.8)^{20-x} \approx 1.38 \times 10^{-8}$$

it seems that this is not a very good estimate.

3.

$$\text{Given } p = P(H) = \frac{1}{2}, \quad q = 1-p = \frac{1}{2}, \quad n = 100$$

$$X : \text{No. of heads}, \quad E(X) = np = 100 \times \frac{1}{2} = 50$$

$$V(X) = npq = 100 \times \frac{1}{2} \times \frac{1}{2} = 25$$

$$P(|X-50| \geq 10) \leq \frac{V(X)}{(10)^2}$$

$$P((X-50) \geq 10 \cup (X-50) \leq -10) \leq \frac{25}{100} = \frac{1}{4}$$

$$P(X \geq 60 \cup X \leq 40) \leq \frac{1}{4} \quad \underline{\text{Ans}}$$

4.

$$p = P(H) = 0.48, \quad q = 1-p = 0.52$$

$$X_i = \begin{cases} 1 & \text{with prob } p \\ 0 & \text{with prob } (1-p) \end{cases}$$

Let there be n toss, the total No. of heads in n toss is $\sum_{i=1}^n X_i$

proportion of heads $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$

$$\mu = E(\bar{X}) = \frac{\sum_{i=1}^n E(X_i)}{n} = \frac{np}{n} = p$$

because $E(X_i) = 1 \cdot p + 0 \cdot (1-p) = p$

$$\sigma^2 = V(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n V(X_i) = \frac{\sum_{i=1}^n p(1-p)}{n^2} = \frac{np(1-p)}{n^2}$$

$$= \frac{p(1-p)}{n}$$

$$E(X_i^2) = 1^2 \cdot p + 0^2 \cdot (1-p) = p$$

$$V(X_i) = E(X_i^2) - (E(X_i))^2$$

$$= p - p^2 = p(1-p)$$

Chebyshev's inequality

$$P(|\bar{X} - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

$$\text{or } P(|\bar{X} - \mu| < \epsilon) \geq 1 - \frac{\sigma^2}{\epsilon^2}$$

$$P(|\bar{X} - 0.48| < 0.01) \geq 1 - \frac{V(\bar{X})}{\epsilon^2} \quad \text{--- (*)}$$

$$\text{Given } P(|\bar{X} - 0.48| < 0.01) \geq 0.95 \quad \text{--- (**)}$$

Compare (*) + (**) we have

$$1 - \frac{V(\bar{X})}{\epsilon^2} = 0.95$$

$$\text{or } \frac{V(\bar{X})}{\epsilon^2} = 0.05$$

$$\frac{p(1-p)}{n \epsilon^2} = 0.05$$

$$n = \frac{p(1-p)}{0.05 \times \epsilon^2} = \frac{0.48 \times 0.52}{(0.05) \times (0.01)^2}$$

$$= \frac{0.2496}{0.00005} = 49920$$

$$\boxed{n = 49920}$$

Q.No.5

Let $X_1, X_2, \dots, X_n$ be i.i.d. $U(0,1)$. Show that $Z_n \xrightarrow{P} C$ where C is a constant. Find C . Where $Z_n = \frac{X_1^2 + X_2^2 + \dots + X_n^2}{n}$

$$f(x) = 1, \quad 0 < x < 1$$

$$E(X_i) = \frac{1}{2}, \quad V(X_i) = \frac{1}{12} \quad \forall i=1, 2, \dots, n.$$

$$E(X_i^2) = V(X_i) + (E(X_i))^2 = \frac{1}{12} + \left(\frac{1}{2}\right)^2 = \frac{1}{12} + \frac{1}{4} = \frac{1+3}{12} = \frac{4}{12} = \frac{1}{3}$$

Using Khinchin's W.L.L.

$$\frac{X_1^2 + X_2^2 + \dots + X_n^2}{n} \xrightarrow{P} E(X_i^2), \text{ as } n \rightarrow \infty$$

$$\Rightarrow Z_n \rightarrow E(X_i^2) = C \text{ as } n \rightarrow \infty$$

$$\text{Hence } C = E(X_i^2) = \frac{1}{3} \text{ Ans.}$$

Q.No.6

$$S_{100} = \sum_{i=1}^{100} X_i$$

X_i denotes the number generated at i th run

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$X \sim U(0,1), \quad E(X) = \frac{1}{2}, \quad V(X) = \frac{1}{12}$$

$$P(S_{100} \geq 52)$$

using CLT

$$P\left(\frac{S_{100} - E(S_{100})}{\sqrt{V(S_{100})}} \geq \frac{52 - 50}{\sqrt{100 \times \frac{1}{12}}}\right)$$

$$\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i = \frac{S_{100}}{100}$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad n=100$$

$$P\left(\frac{\frac{S_{100}}{100} - \frac{1}{2}}{\sqrt{\frac{1}{12 \times 100}}} \geq \frac{\frac{52}{100} - \frac{1}{2}}{\sqrt{\frac{1}{12 \times 100}}}\right)$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$P\left(Z \geq \frac{(0.52 - 0.5) \times 10 \sqrt{12}}{1}\right)$$

$$P(Z \geq 0.69) = 0.5 - P(0 < Z < 0.69) = 0.5 - 0.2549 = 0.2451 \text{ Ans.}$$

X : Cholesterol of a person over age 50

$$E(X) = 54, \quad V(X) = (17)^2, \quad \sigma = 17$$

$$\bar{X} = \frac{1}{40} \sum_{i=1}^{40} X_i$$

using CLT $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$$P(\bar{X} \geq 60)$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{60 - 54}{17/\sqrt{40}}\right) = P\left(Z \geq \frac{60 - 54}{17/\sqrt{40}}\right)$$

$$P\left(Z \geq \frac{6\sqrt{40}}{17}\right) = P(Z \geq 2.23)$$

$$= 0.5 - P(0 \leq Z \leq 2.23) = 0.5 - 0.4871$$

$$= 0.0129$$

Q.No. 8

$$X_j \sim N(0,1)$$

$$X_j \stackrel{iid}{\sim} N(0,1); \quad j=1, 2, \dots, n$$

$$Y_n = \frac{1}{n} \sum_{j=1}^n X_j^2$$

$$E(X_j^2) = V(X_j) + (E(X_j))^2$$

$$= 1 + 0 = 1$$

using CLT $P(Y_n \geq 1.01) = 0.05$

$$X_j^2 \sim \chi_1^2, \text{ Hence, } V(X_j^2) = 2$$

$$P\left(\frac{Y_n - 1}{\sqrt{2/n}} \geq \frac{1.01 - 1}{\sqrt{2/n}}\right) = 0.05$$

$$E(Y_n) = \frac{1}{n} \sum_{j=1}^n E(X_j^2)$$

$$= 1$$

$$V(Y_n) = \frac{1}{n^2} \sum_{j=1}^n V(X_j^2)$$

$$P\left(Z \geq \frac{0.01\sqrt{n}}{\sqrt{2}}\right) = 0.05$$

$$= \frac{2}{n}$$

From Normal table

$$\frac{0.01\sqrt{n}}{\sqrt{2}} = 1.65$$

$$\frac{\sqrt{n}}{\sqrt{2}} = \frac{1.65}{0.01}$$

$$n = \frac{(1.65)^2 \times 2}{(0.01)^2} = \frac{5.445}{0.0001} = \underline{\underline{54450}}$$

Q.No. 9 (a)

$$E(X) = \mu = \int_1^{\infty} \frac{5}{x^5} dx = \frac{5}{4}$$

$$\sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2 = \int_1^{\infty} \frac{5}{x^4} dx - \frac{25}{16} = \frac{5}{48}$$

In order to use Chebychev's, we need to have a symmetric probability, i.e. $P(|X - \mu| \geq r) \leq \frac{V(X)}{r^2}$

$$\Rightarrow P(X \geq 2.5) = P(X \geq \frac{5}{4} + \frac{5}{4}), \text{ so write}$$

~~$$P(|X - \mu| \geq \frac{5}{4}) = P(X \geq \frac{5}{4}) + P(X \leq \frac{5}{4})$$~~

~~$$\Rightarrow P(|X - \frac{5}{4}| \geq 2.5) = P((X - 1.25) \geq 2.5)$$~~

~~$$\text{or } (X - 1.25) \leq -2.5$$~~

$$\Rightarrow P(|X - \frac{5}{4}| \geq \frac{5}{4}) = P((X - \frac{5}{4}) \geq \frac{5}{4})$$

$$\text{or } (X - \frac{5}{4}) \leq -\frac{5}{4}$$

$$= P(X \geq 2.5 \text{ or } X \leq 0)$$

$$= P(X \geq 2.5) \leq \frac{\sigma^2}{(5/4)^2} = \frac{5/48}{25/16} = \frac{1}{15} \text{ Ans}$$

If we want $P(X \geq \alpha) \leq 15\%$, using Chebychev's inequality, then we need

$$\frac{15}{100} = \frac{\sigma^2}{r^2} = \frac{5/48}{r^2}$$

$$r = \sqrt{\frac{5/48}{15/100}} = \sqrt{\frac{100}{3 \times 48}} = \frac{10}{12} = \frac{5}{6}$$

$$\therefore P(X \geq \mu + 5/6) \text{ and } \mu = \frac{5}{4} + \frac{5}{6} = \frac{25}{12}$$

Ans

(b)

$$B_n = V(X_1 + X_2 + \dots + X_n)$$

$$= \sum_{i=1}^n V(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)$$

$$\text{Cov}(X_i, X_j) = -\frac{\sigma^2}{n-1}$$

$$= n\sigma^2 + n(n-1)\left(-\frac{\sigma^2}{n-1}\right) = n\sigma^2 - n\sigma^2 = 0$$

$$\frac{B_n}{n^2} = \frac{0}{n^2} = 0$$

Hence $\{X_n\}$ holds the law of large number.
* n.